

Optimization

Thursday, August 20, 2026

9:04 AM

Unconstrained

$$\max_{x \in D} f(x)$$

First order

necessary
condition

$$\nabla f(x^*) = 0$$

for interior
solution

Second order
condition

$$\nabla^2 f(x^*) = H_f \succeq 0$$

if f strictly concave

First order condition sufficient
for global max

if not, check list of $\{x_i^*\}$

$f(x_i^*)$ that passed 1st & 2nd

as well as boundary points

What boundary
points if
unconstrained?

Constrained optimization

Constrained optimization

$$\max f(x) \text{ s.t. } g(x) = 0$$

$$g(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix} = 0$$

Lagrangian

$$L(x, \lambda) = f(x) + \lambda^T g(x)$$

First order condition

$$\nabla_x L(x, \lambda) = 0$$

$$= \underbrace{\nabla_x f(x)}_{n \times 1} + \underbrace{D_x g(x)^T}_{n \times m} \underbrace{\lambda}_{m \times 1} = 0$$

2nd order condition

$$D_x^2 L(x, \lambda) \preceq 0 \quad \text{neg. semi-definite}$$

Constant qualification

λ encoding of g constants
... .. but we

a necessary condition
holds locally but we
don't want constraints
to globally collapse

check that $\{ \nabla g_i(x) \}_{i=1}^m$
is linearly independent

Finally, check critical points
that passed above
for points on boundary

BUT if boundary points are
candidate maximizers,
shouldn't we use
inequality bounds?

Equality bounds assume
 f & g functions
are continuously differentiable

Ex:

$$\max_{x \in \mathbb{R}} f(x) = x \quad \text{s.t.} \quad x^2 = 0$$

Lagrangian fails

$$L(x, \lambda) = x + \lambda x^2$$

$$\text{FOL} \quad \frac{\partial}{\partial x} L(x, \lambda) = 1 + 2\lambda x = 0$$
$$x = \frac{1}{2\lambda}$$

breaks $x^2 = 0$

so theory speaking,
what about this example
breaks Lagrangian process?
what are examples
of constraints not
being linearly independent?

Inequality constraints

$$\max_{x \in \mathbb{R}} f(x) \quad \text{s.t.} \quad g(x) \geq 0$$

(a.k.a.)

$$x \in D \quad f(x) = \dots \quad \bigcup_{i=1}^m \{x \mid g_i(x) \geq 0\}$$

$$\begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix} \geq 0$$

$$f(x) = 0$$

$$\begin{bmatrix} h_1(x) \\ \vdots \\ h_l(x) \end{bmatrix} = 0$$

1) Lagrangian

$$L(x, \lambda, \mu) =$$

$$f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^l \mu_j h_j(x)$$

$$\text{w/ } \lambda_i, \mu_j \geq 0$$

Note: if $g(x) \geq 0$

then we use

$$+ \sum \lambda_i g_i(x) \text{ w/ } \lambda_i \geq 0$$

sometimes economists use

$$g(x) \leq 0$$

$$+ - \sum \lambda_i g_i(x)$$

$$\text{w/ } \lambda_i \geq 0$$

Karush Kuhn Tucker (KKT)
conditions

first order conditions

$$\nabla f(x^*) + \sum_i^m \lambda_i \nabla g_i(x^*) + \sum_j \mu_j \nabla h_j(x^*) = 0$$

w/ feasibility conditions

$$g_i(x^*) \leq 0 \text{ \& } h_j(x^*) = 0$$

$$\lambda_i^*, \mu_j^* \geq 0$$

primal
vs
dual?

complementary slackness

$$\lambda_i^* g_i(x^*) = 0$$

$\Rightarrow$ either $\lambda_i^* = 0$

or $g_i(x^*) = 0$

constraint only binds

if $\lambda_i^* > 0$

λ s are
shadow

prices

in micro

models

both λ_i & $g_i(x)$

could be zero

but then it's

trivial constraint?

Envelope Theorem

max problem w/ parameter

$$\max_x f(x, \theta) = V(\theta)$$

value function
that depends on
environment

if maximizer $x^*(\theta)$ exists
if environment θ changes,
 x^* can change

$$x^*(\theta) \in \arg \max_x f(x, \theta)$$

$$V(\theta) = f(x^*(\theta), \theta)$$

$$V'(\theta) = \frac{dV}{d\theta} = f_1(x^*(\theta), \theta) \frac{dx^*}{d\theta} + f_2(x^*(\theta), \theta)$$

by optimal path, $f_1 \frac{dx^*}{d\theta} = 0$

so value $V'(\theta) = f_2(x^*(\theta), \theta)$

$$= \left. \frac{df(x, \theta)}{d\theta} \right|_{x=x^*(\theta)}$$

parameters use implicit functions
& chain rule

change in value function can
come from either choices x
or environment θ (ind response
in x as result of θ change)

but first order conditions
assume $f_1 = 0$ why?
example?

for constrained problem:
use Lagrangian but include θ

$$L(x, \lambda, \mu, \theta)$$

$$\text{s.t. } \frac{\partial L(x^*(\theta), \lambda^*(\theta), \mu^*(\theta), \theta)}{\partial \theta} \bigg|_{x^*, \lambda^*, \mu^*}$$

Envelope theorem is shortcut
that uses value function

that uses value function
to show in first order,
something? does not
matter

you could calculate full optimization
but envelope saves steps

Not covering: when is value
function differentiable?
used for dynamic
programming

Likewise, correspondences will
be covered later in math methods

Maximum theorem extends continuity
of value function w/ set
value mapping

is value function w/ set
value mapping

fixed point theorem also
comes later

Example from 2025 exam:

$$\max_{x, y, z \in \mathbb{R}^3} f(x) = -x^2 + 2x - y^2 + 4y - z^2 + 6z$$

$$\text{s.t. } x \leq 2, y + 2z \geq 12$$

for rewrite constraints

$$-x + 2 \geq 0$$

$$y + 2z - 12 \geq 0$$

$$\mathcal{L}(x, y, z, \mu_1, \mu_2) =$$

$$-x^2 + 2x - y^2 + 4y - z^2 + 6z$$

$$+ \mu_1(-x + 2) + \mu_2(y + 2z - 12)$$

$$\mu_1, \mu_2 \geq 0$$

F.O.C.:

$$\frac{\partial Z}{\partial x} = -2x + 2 - \mu_1 = 0$$

$$\frac{\partial Z}{\partial y} = -2y + 4 + \mu_2 = 0$$

$$\frac{\partial Z}{\partial z} = -2z + 6 + 2\mu_2 = 0$$

compl. slackness

$$\mu_1(-x+2) = 0$$

$$\mu_2(y+2z-12) = 0$$

Feasibility:

$$-x+2 \geq 0$$

$$y+2z-12 \geq 0$$

$$\mu_1, \mu_2 \geq 0$$

if $x=2$,

but by $\frac{\partial Z}{\partial x}$, $\mu_1 = -2x+2 = -2$
but then $\mu_1 < 0$,
breaking feasibility.

so $\mu_1^* = 0$ must hold

$$\text{so by F.O.C., } -2x+2=0 \\ x^* = 1$$

suppose $y+2z-12 > 0$,
c.t. $\mu_2 = 0$

suppose $y + 2z = 12$,
s.t. $\mu_2 = 0$,

then $y = 2$ & $z = 3$
are infeasible

$$\text{by } y + 2z - 12 \geq 0$$

$2 + 6 - 12 = -4 < 0$
contradiction so

$$y + 2z - 12 = 0$$

must hold

$$\text{so } y + 2z - 12 = 0$$

$$\text{+ } 4y - 2z - 2 = 0 \text{ \& where is this from?}$$

$$\text{from } -2y + 1 + \mu_2 = 0 = -2z + 6 + 2\mu_2$$

$$\text{solve for } y^* = \frac{14}{5}, z^* = \frac{23}{5}$$

$$\text{check } \mu_2^* = \frac{8}{5} > 0 \text{ is feasible}$$

if we arrive at non feasible
solution, then problem may not
be feasible

Did not
need
second
order
since cost
path was
unique

be feasible

image

we can also have multiple solutions if feasibility doesn't eliminate enough arguments, then evaluate function values to pick max

is there a matrix way to solve Lagrangians w/ inequalities?

Ensure constraints are stated correctly in form

$$g_i = X_i + y_i \geq 0$$

greater than or
weakly